

LAM Research C++ Preparation - Session 1 Summary

Goal:

Build C++ foundation for LAM Research Software Engineer interview, DSA in C++, and low-level understanding.

1. Why C++ over Python?

- C++ is preferred for performance-critical systems.
- It provides low-level memory control, less runtime overhead, and efficient execution.
- Used in OS, embedded systems, semiconductor software, competitive programming.

2. C++ Execution Flow:

C++ Source Code -> Compiler -> Machine Code -> OS loads process -> CPU executes.

3. Basic C++:

`#include` provides `cin` and `cout`.

`main()` is the execution starting point.

`return 0` indicates successful completion.

4. Memory Layout:

Code Segment: Program instructions.

Data Segment: Global and static variables.

Stack: Local variables, parameters, function calls.

Heap: Dynamic memory allocation.

5. Stack Memory:

- Fast
- Automatically managed
- Contiguous memory
- Limited size

6. Heap Memory:

- Dynamic allocation
- Larger memory
- Slower due to allocation overhead and fragmentation.

7. Static Variables:

- Created once.
- Retain values between function calls.
- Stored in data segment.

8. Pointers:

A pointer stores an address.

Example:

```
int x = 10;
```

```
int *ptr = &x;
```

& gives address.

* accesses value at address.

9. Dynamic Memory:

```
int *ptr = new int(50);
```

ptr is stored in stack.

50 is stored in heap.

10. delete and delete[]:

new -> delete

new[] -> delete[]

11. Memory Leak:

Allocated memory not released.

Causes increasing RAM usage and performance issues.

12. Segmentation Fault:

Occurs when a program accesses invalid memory.

Examples:

- Dereferencing nullptr
- Accessing invalid array index
- Using deleted memory

13. Pass by Value:

Creates a copy.

Original data is unchanged.

14. Pass by Reference:

Uses the same memory.

Example:

```
void solve(vector& nums)
```

Advantages:

- Avoids copying
- Improves performance

15. Pointer vs Reference:

Pointer stores an address and can be changed.

Reference is another name for an existing variable.

16. Next Topics:

- C++ OOPS
- Class and Object
- Constructor
- Destructor
- Encapsulation
- Inheritance
- Polymorphism
- Virtual Functions
- STL for DSA
- Deep revision of pointers and references

LAM Preparation Progress:

Completed:

- Memory concepts
- Stack vs Heap
- Dynamic allocation basics
- Pointer basics
- References introduction

Need revision:

- Pointers
- References
- STL syntax